



Secrets of the secret lunar station

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Game dev with 10+ years in R&D as a mechanical
design engineer. Proudly led the team behind
Thunder Age, a post-apocalyptic tower defense! ⚡





Games Goals

- SOTSLS is mainly an FPS shooter with waves of mobs that you need to defeat.
- Heat gun system
- It has an interesting (Charge Undo System)
- Its hard and pretty short

Existing Data and or Algorithms

- Spawner System, Gun Heat system and flashback or charge undo system

[Changed From Array to List and Changed the Spawner · KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked@d100e2d · GitHub](#)





Changed the EnemySpawner from array to list

- If we were to use an array we would have to determine the fixed size of enemies. No Fixed size needed with list
- We can also add or remove enemies dynamically during gameplay without resizing.

- [Changed From Array to List and Changed the Spawner - KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked@d100e2d · GitHub](https://github.com/KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked@d100e2d)





LinkedList for the UNDO system

- More Flexibility
- Efficient insertions and deletions
- Better for Expanding Undo Systems
- <https://github.com/KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked/commit/0110e21f5c1ab279a15d014b6d8a64e0fa05bba0>

New Algorithm

- So we wanted to make the game easier and give a fun challenge.
- We decided to add a difficulty system. By pressing H it increases the amount of Enemies that spawn
- By using a switch case we are able to increment the amount of MaxEnemies that spawn and augment the spawn speed
- Difficulty for game with letter H ·
[KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked@0a2a0ff · GitHub](#)



Deleting Useless Code

- For some reason the developer made the same script twice and never deleted the original when he made changes
- <https://github.com/KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked/commit/0a2a0ff62eea89f392639a5955c048cf03ab1a69>

```
... @@ -1,39 +0,0 @@
1  - using System.Collections;
2  - using System.Collections.Generic;
3  - using UnityEngine;
4  -
5  - //public class GunHeat : MonoBehaviour
6  - //{
7  -     //[SerializeField]
8  -     // Renderer[] heatLevel;
9  -     //[SerializeField]
10 -     // Material coolMaterial;
11 -     //[SerializeField]
12 -     // Material hotMaterial;
13 -     //[SerializeField]
14 -     // ShootingScript shootingScript;
15 -
16 -
17 -
18 -     // Start is called before the first frame update
19 -     // void Start()
20 -     // {
21 -
22 -     // }
23 -
24 -     // Update is called once per frame
25 -     // void Update()
26 -     // {
27 -     //another thing we can edit
28 -     //     if (shootingScript.ShotsFired > 19) heatLevel[0].material = hotMaterial;
29 -     //     else heatLevel[0].material = coolMaterial;
30 -     //     if (shootingScript.ShotsFired > 39) heatLevel[1].material = hotMaterial;
31 -     //     else heatLevel[1].material = coolMaterial;
32 -     //     if (shootingScript.ShotsFired > 59) heatLevel[2].material = hotMaterial;
33 -     //     else heatLevel[2].material = coolMaterial;
34 -     //     if (shootingScript.ShotsFired > 79) heatLevel[3].material = hotMaterial;
35 -     //     else heatLevel[3].material = coolMaterial;
36 -     //     if (shootingScript.ShotsFired > 99) heatLevel[4].material = hotMaterial;
37 -     //     else heatLevel[4].material = coolMaterial;
38 -     // }
39 - //}
```

Bug Fix

- Fixed a bug we made relating to the enemy spawn system
 - Enemies kept spawning after difficulty increase
- <https://github.com/KyleBermundo/Secrets-of-the-Secret-Lunar-Station-Forked/commit/0a2a0ff62eea89f392639a5955c048cf03ab1a69>

Conclusion

- Files were a bit unorganized or were just way too many cs files
- There weren't many things we could edit
- Difficult when testing base game

Questions?